



1. Description

1.1. Project

Project Name	g4-oscilloscope-b
Board Name	NUCLEO-G474RE
Generated with:	STM32CubeMX 6.17.0
Date	06/14/2026

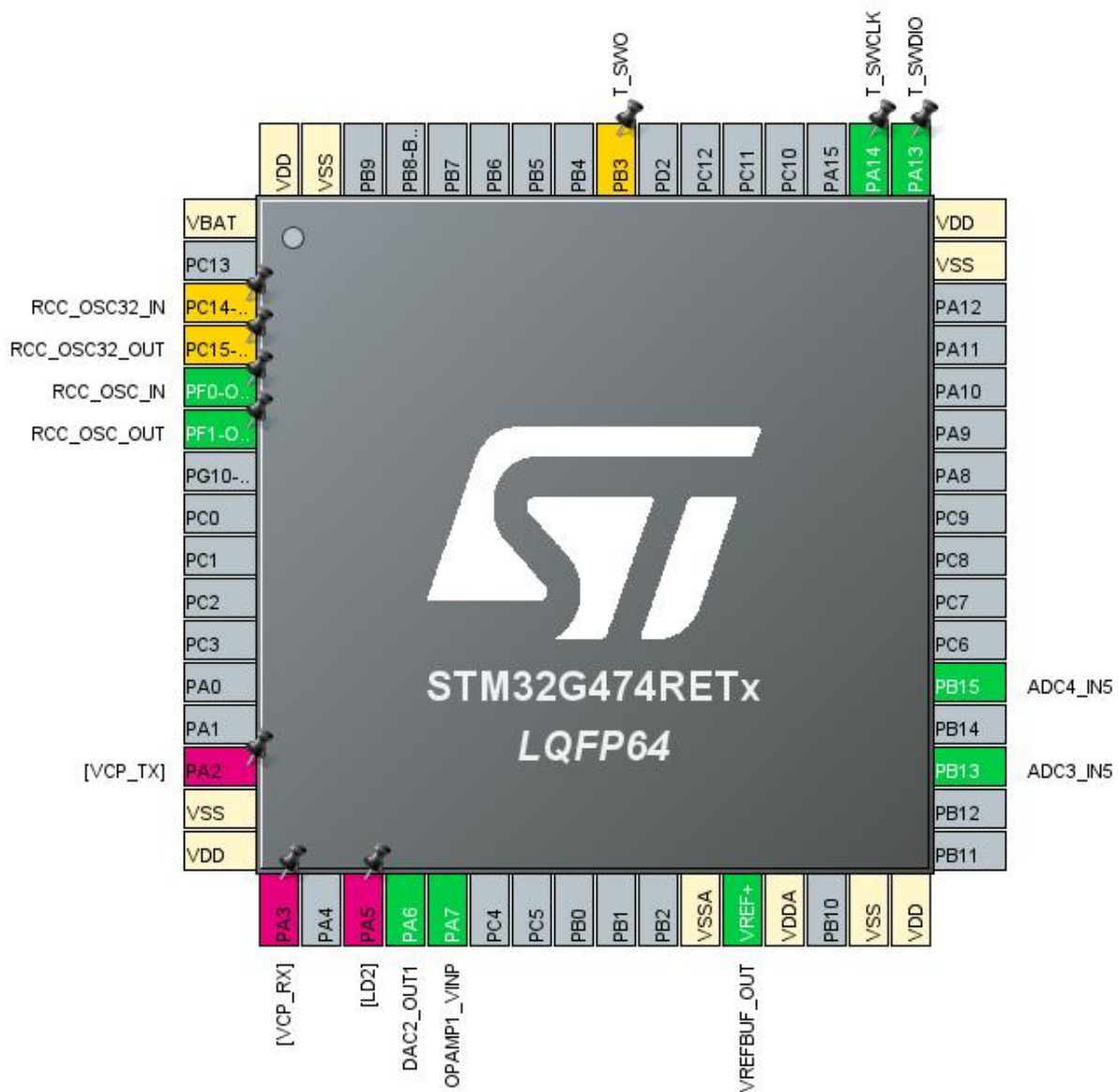
1.2. MCU

MCU Series	STM32G4
MCU Line	STM32G4x4
MCU name	STM32G474RETx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration



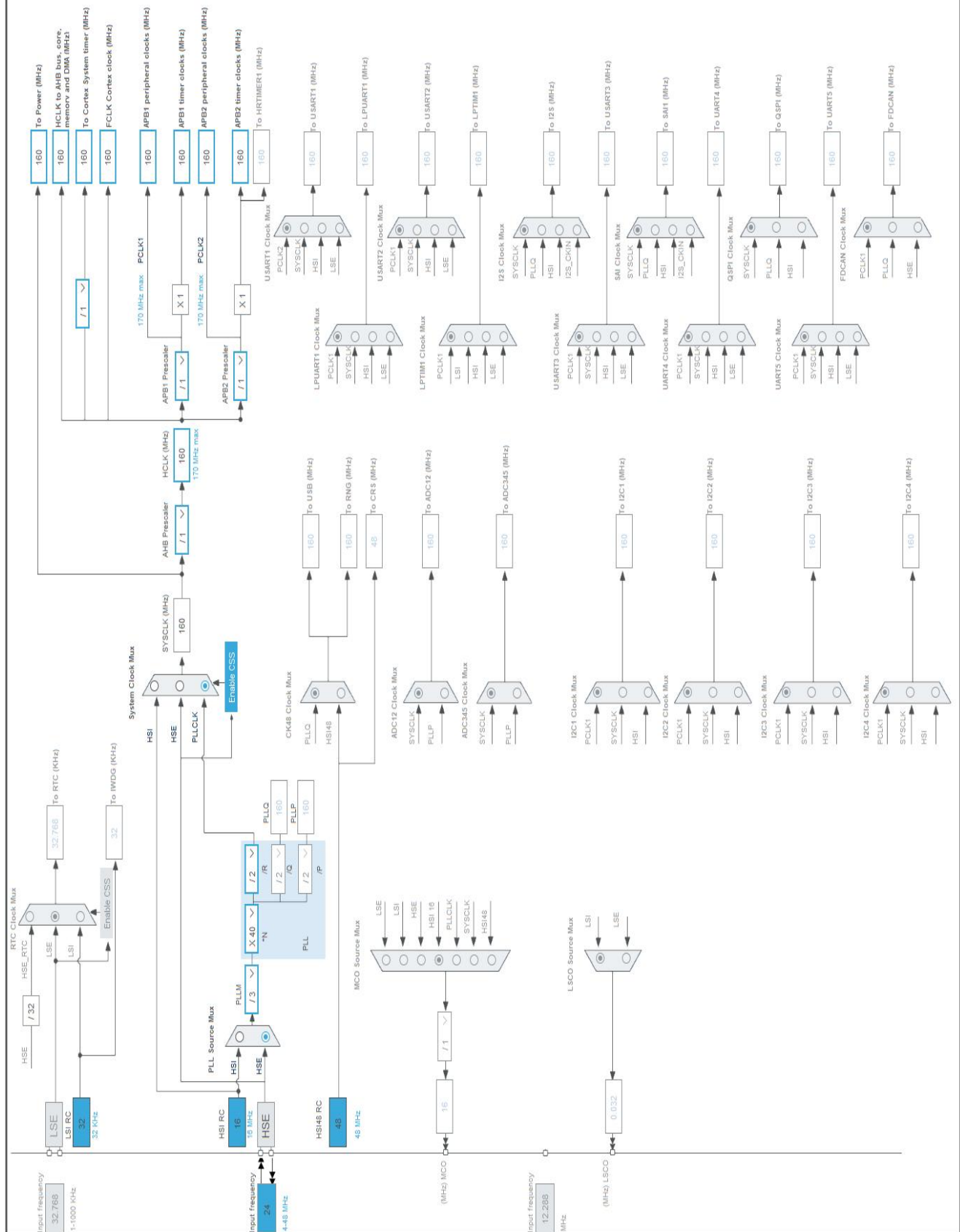
3. Pins Configuration

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
3	PC14-OSC32_IN *	I/O	RCC_OSC32_IN	RCC_OSC32_IN
4	PC15-OSC32_OUT *	I/O	RCC_OSC32_OUT	RCC_OSC32_OUT
5	PF0-OSC_IN	I/O	RCC_OSC_IN	RCC_OSC_IN
6	PF1-OSC_OUT	I/O	RCC_OSC_OUT	RCC_OSC_OUT
14	PA2	I/O		
15	VSS	Power		
16	VDD	Power		
17	PA3	I/O		
19	PA5	I/O		
20	PA6	I/O	DAC2_OUT1	
21	PA7	I/O	OPAMP1_VINP	
27	VSSA	Power		
28	VREF+	MonoIO	VREFBUF_OUT	
29	VDDA	Power		
31	VSS	Power		
32	VDD	Power		
35	PB13	I/O	ADC3_IN5	
37	PB15	I/O	ADC4_IN5	
47	VSS	Power		
48	VDD	Power		
49	PA13	I/O	SYS_JTMS-SWDIO	T_SWDIO
50	PA14	I/O	SYS_JTCK-SWCLK	T_SWCLK
56	PB3 *	I/O	SYS_JTDO-SWO	T_SWO
63	VSS	Power		
64	VDD	Power		

* The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration

RIF configuration not available for STM32G474RETx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32G4
Line	STM32G4x4
MCU	STM32G474RETx
Datasheet	DS12288 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

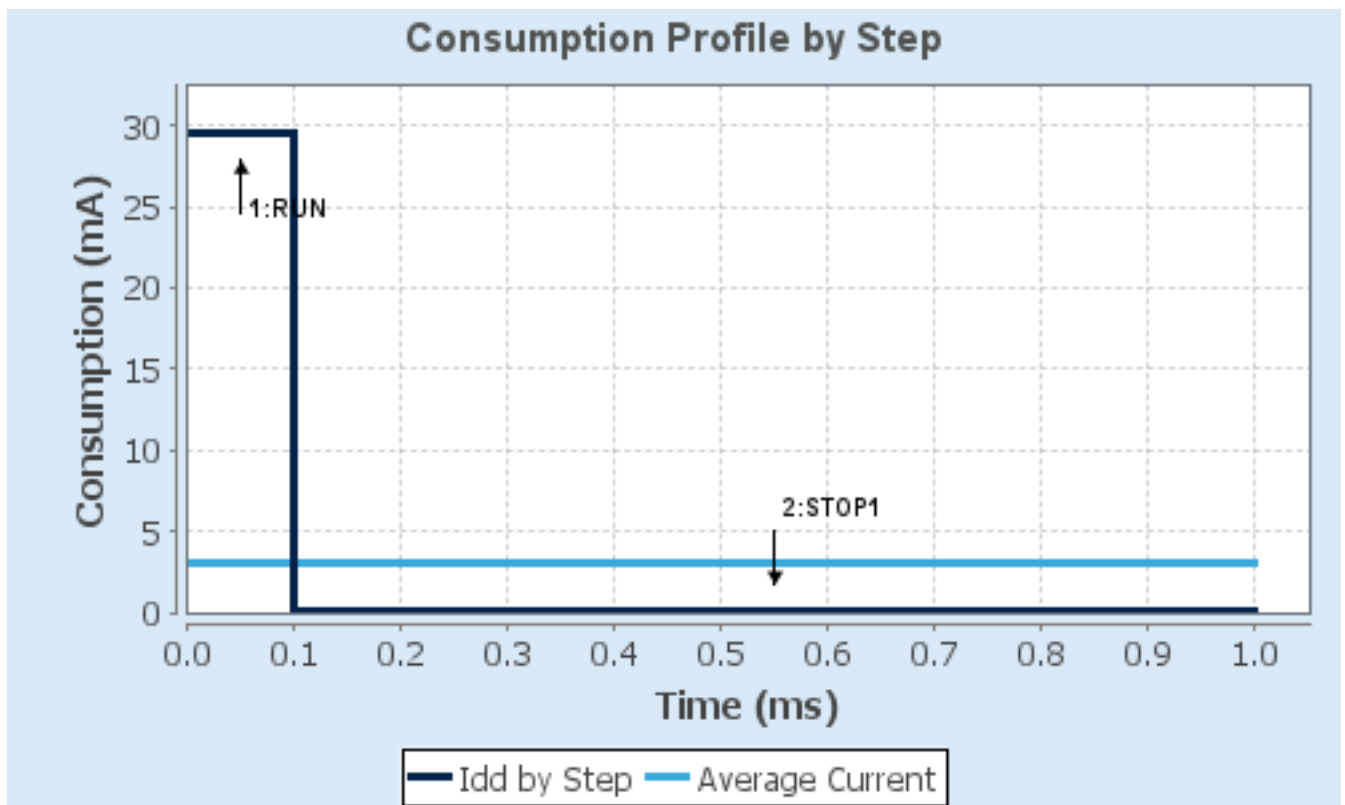
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP1
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Boost	NoRange
Fetch Type	FLASH/DualBank/ART	NA
CPU Frequency	170 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	29.5 mA	80.5 μ A
Duration	0.1 ms	0.9 ms
DMIPS	213.0	0.0
Ta Max	124.25	129.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.02 mA
Battery Life	1 month, 16 days, 9 hours	Average DMIPS	212.5 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	g4-oscilloscope-b
Project Folder	D:\projects\g4-oscilloscope-b
Toolchain / IDE	CMake
Firmware Package Name and Version	STM32Cube FW_G4 V1.6.3
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_ADC1_Init	ADC1
4	MX_ADC3_Init	ADC3
5	MX_ADC4_Init	ADC4
6	MX_OPAMP1_Init	OPAMP1
7	MX_TIM2_Init	TIM2
8	MX_DAC2_Init	DAC2

3. Peripherals and Middlewares Configuration

3.1. ADC1

mode: VOPAMP1 Channel

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler Synchronous clock mode divided by 4

Resolution ADC 12-bit resolution

Data Alignment Right alignment

Gain Compensation 0

Scan Conversion Mode Disabled

End Of Conversion Selection End of single conversion

Low Power Auto Wait Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

DMA Continuous Requests Disabled

Overrun behaviour Overrun data preserved

ADC_Regular_ConversionMode:

Enable Regular Conversions Enable

Enable Regular Oversampling Disable

Number Of Conversion 1

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel Channel Vopamp1

Sampling Time 2.5 Cycles

Offset Number No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode false

3.2. ADC3

mode: IN5

3.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode

Dual interleaved mode only *

DMA Access Mode

DMA access mode disabled

Delay between 2 sampling phases

1 Cycle

ADC_Settings:

Clock Prescaler

Synchronous clock mode divided by 4

Resolution

ADC 12-bit resolution

Data Alignment

Right alignment

Gain Compensation

0

Scan Conversion Mode

Disabled

End Of Conversion Selection

End of single conversion

Low Power Auto Wait

Disabled

Continuous Conversion Mode

Disabled

Discontinuous Conversion Mode

Disabled

DMA Continuous Requests

Disabled

Overrun behaviour

Overrun data preserved

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Enable Regular Oversampling

Disable

Number Of Conversion

1

External Trigger Conversion Source

Regular Conversion launched by software

External Trigger Conversion Edge

None

Rank

1

Channel

Channel 5

Sampling Time

2.5 Cycles

Offset Number

No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions

Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode

false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode

false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode

false

3.3. ADC4

mode: IN5

3.3.1. Parameter Settings:

ADCs_Common_Settings:

Mode

Dual interleaved mode only *

DMA Access Mode

DMA access mode disabled

Delay between 2 sampling phases

1 Cycle

ADC_Settings:

Clock Prescaler

Synchronous clock mode divided by 4

Resolution

ADC 12-bit resolution

Data Alignment

Right alignment

Gain Compensation

0

Scan Conversion Mode

Disabled

End Of Conversion Selection

End of single conversion

Low Power Auto Wait

Disabled

Continuous Conversion Mode

Disabled

Discontinuous Conversion Mode

Disabled

DMA Continuous Requests

Disabled

Overrun behaviour

Overrun data preserved

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Enable Regular Oversampling

Disable

Number Of Conversion

1

Rank

1

Channel

Channel 5

Sampling Time

2.5 Cycles

Offset Number

No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions

Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode

false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode

false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode

false

3.4. DAC2

OUT1 mode: Connected to external pin and to on chip-peripherals

3.4.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	Timer 2 Trigger Out event *
Trigger2	None
Wave generation mode	Triangle wave generation *
Maximum Triangle Amplitude	1023 *
User Trimming	Factory trimming

3.5. NUCLEO-G474RE

mode: Human Machine Interface

3.5.1. Human Machine Interface:

Led:

USER LED GREEN (LD2) **true ***

Button:

USER B1 Disable

VCOM:

Virtual Com Port **true ***

Demonstration code:

Generate demonstration code Disabled

3.6. OPAMP1

Mode: PGA Internally Connected

3.6.1. Parameter Settings:

Basic Parameters:

Power Mode	Normal
PGA Gain	2 or -1
User Trimming	Disable

3.7. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.7.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	4 WS (5 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1 boost
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Peripherals Clock Configuration:

Generate the peripherals clock configuration	TRUE
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3.8. SYS

Debug: Serial Wire

VREFBUF Mode: Internal voltage reference

Timebase Source: SysTick

mode: save power of non-active UCPD - deactive Dead Battery pull-up

3.8.1. Parameter Settings:

Voltage_Reference_Buffer_Settings:

Trimming Mode	Factory Trimming
Internal Voltage reference scale	SCALE 0: around 2.048 V

3.9. TIM2

Clock Source : Internal Clock

3.9.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	9 *
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	9 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Update Event *

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC3	PB13	ADC3_IN5	Analog mode	No pull-up and no pull-down	n/a	
ADC4	PB15	ADC4_IN5	Analog mode	No pull-up and no pull-down	n/a	
DAC2	PA6	DAC2_OUT1	Analog mode	No pull-up and no pull-down	n/a	
OPAMP1	PA7	OPAMP1_VINP	Analog mode	No pull-up and no pull-down	n/a	
RCC	PF0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	RCC_OSC_IN
	PF1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	RCC_OSC_OUT
SYS	VREF+	VREFBUF_OUT	n/a	n/a	n/a	
	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	T_SWDIO
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	T_SWCLK
Single Mapped Signals	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	RCC_OSC32_IN
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	RCC_OSC32_OUT
	PB3	SYS_JTDO-SWO	n/a	n/a	n/a	T_SWO

4.2. DMA configuration

nothing configured in DMA service

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	0	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1 and ADC2 global interrupt	unused		
TIM2 global interrupt	unused		
ADC3 global interrupt	unused		
TIM7 global interrupt, DAC2 and DAC4 channel underrun error interrupts	unused		
ADC4 global interrupt	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Middleware								
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Utilities	Bsp
DMA	ADC1 ✓	TIM2 ✓						NUCLEO-G474RE ✓
GPIO ⚠	ADC3 ✓							
I2C ✓	ADC4 ✓							
RCC ✓	DAC2 ✓							
SYS ✓	OPAMP1 ✓							

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32g4_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32g4_svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-usb-c-pd-solutions-presentation.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf
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Flyers	https://www.st.com/resource/en/flyer/flstm32g4.pdf
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